



Intents in Android

One factor contributing to Android's success in the smartphone market is the large number of third-party application developers in Java. The other not-so-well-known yet important factor is the ease with which developers can write complex applications with Android's very good inter-application communication mechanism (IACM), facilitating component re-use and faster turn-around while writing new applications. This article explores 'intents'—the heart of Android's IACM.

First, let's get familiar with the Android terminology used in this article:

- **Activity:** The user interface part of an application. An application can have one or more activities.
- **Service:** Code that runs in the background (typically to perform a long operation) and does not interact with the user.
- **Broadcast Receiver:** Code that registers ('listens') for a particular broadcast message.

What are intents?

In Android, 'intent' describes 'an operation to be performed'. For example, it can say 'open a browser', 'send an SMS', 'make a phone call', etc. In some cases, an intent can also describe something that has happened in the past, and is now being reported to the Android system. Examples of this include reporting that the battery status has changed, a head-set has been plugged in, a camera button was pressed, etc. This category is mostly used by the Android OS to send 'system-wide' notifications about a certain event, and is usually referred to as 'System Broadcast Intents'. These can only be sent by the system, not by an application. Intents can be sent between Activities, Services and Broadcast Receivers. Applications use intents for both inter- and intra-application communication.

The components of an intent

An intent itself is an object of the *Intent* class, which meets the descriptions we talked about earlier. Broadly, the descriptions can be classified as given below:

Component Name (Optional)	Describes what component should handle the intent. It is specified by the fully qualified class name of the component and the package name, with the project name, e.g., "com.example.project.Ify.SampleActivity" with "com.example.project".
Action	Describes what action to be taken; e.g., <code>ACTION_DIAL</code> to specify 'make a phone call'.
Data	Contains data required for the action; e.g., the phone number to call.
Category	Provides additional information about the action to perform; e.g., <code>CATEGORY_LAUNCHER</code> means it should appear in the Launcher as a top-level application.
Extras (Optional)	Key-value pairs holding additional information, if any. Mostly used in the broadcast intents. For example, while reporting battery status, the intent describes battery 'capacity', 'temperature', etc., as extras.